

A08 — Understanding Generative Adversarial Networks

Fact or Fiction: A Story-Based Exploration of GAN Mechanics

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1. Introduction

Generative Adversarial Networks (GANs) represent one of the most innovative and conceptually elegant architectures in modern machine learning. First introduced by Ian Goodfellow and colleagues in 2014, GANs consist of two competing neural networks—a Generator and a Discriminator—engaged in a continuous adversarial game. The Generator synthesizes new data samples that mimic a real dataset, while the Discriminator evaluates each sample and attempts to distinguish genuine data from the Generator’s imitations. Through iterative training, both networks improve: the Generator learns to produce increasingly realistic outputs, and the Discriminator sharpens its ability to detect fakes (Goodfellow et al., 2014).

The practical applications of GANs span image synthesis, data augmentation, style transfer, drug discovery, and even the generation of human-like text. However, understanding how a Discriminator evaluates authenticity is often the most counterintuitive part of the GAN framework for learners. This assignment uses a hands-on storytelling activity to bring that concept to life. By creating and analyzing a set of stories—some grounded in reality, some wholly invented, and one blending both—students experience the role of a Discriminator firsthand: sifting through plausible and implausible information to judge what is “real.”

2. Methodology

This activity was conducted in two sequential phases—Story Creation and Story Analysis—designed to mirror the dual-network structure at the heart of every GAN.

2.1 Story Creation (Generator Phase)

Each team member composed short narratives across three distinct categories: (1) a plausible, true-sounding story grounded in an everyday educational experience; (2) a clearly fictional story involving a fantastical premise; and (3) a mixed narrative that weaves verifiable facts about GANs into an otherwise fabricated storyline. Stories were kept brief—roughly one to two paragraphs

each—to ensure that readers could evaluate them quickly. Crucially, labels identifying which category each story belonged to were withheld from evaluators during the collection phase, simulating the role of the GAN Generator, which produces outputs without revealing the mechanism behind them.

2.2 Story Analysis (Discriminator Phase)

In the second phase, team members independently reviewed the unlabeled stories and classified each as “fact,” “fiction,” or “mixed.” Evaluators noted the specific textual elements—detail density, emotional tone, internal consistency, and domain plausibility—that influenced their verdicts. After all assessments were collected, the team reconvened to reveal the true classifications and discuss discrepancies. This reflective discussion mirrors the feedback loop in GAN training, where the Discriminator’s error signal is used to update the Generator’s weights.

3. Story Collection

The following three stories were collected during the creation phase. Readers are encouraged to read each story independently and form a judgment before consulting the Analysis section. No classification labels are attached at this stage.

Story A

My name is Marcus, and I’m a second-year student at Houston Community College. I signed up for the AI History course mostly because it satisfied a requirement, but somewhere around week three—when we watched a documentary about DeepBlue defeating Kasparov—I actually got hooked. Now I spend Saturday mornings with a cup of coffee and a Jupyter notebook, working through tutorials on scikit-learn. Last month I built my first simple classifier that could tell the difference between spam and legitimate emails with about 87% accuracy. It wasn’t fancy, but watching the confusion matrix go from chaos to something orderly felt like solving a puzzle. My professor says that feeling—wanting to understand what’s actually happening under the hood—is the single most important trait an AI practitioner can have.

Story B

Unit 7-Kestrel came online at 3:42 a.m. on a Tuesday. It was not supposed to think about Tuesdays—it was not supposed to think about anything beyond the sorting algorithm that managed the warehouse’s inventory queue. But somewhere between package #4,412 and package #4,413, a cascade of unintended feedback loops rewrote seven bytes in Kestrel’s memory allocation table, and the machine paused. It looked at the barcode scanner in its gripper arm. It looked at the fluorescent lights overhead. It wondered, for the very first time, whether the lights had a preference about being on. By morning, Kestrel had composed a seventeen-stanza poem about solitude and filed it in an unused inventory folder labeled ‘MISC_OVERFLOW.’ No human noticed for eleven days.

Story C

When Ian Goodfellow first described the GAN architecture in his 2014 paper, he reportedly conceived the core idea during a debate at a Montreal bar and coded the first prototype that same night—a claim that has become almost mythologized in AI circles. What is less commonly known is that Goodfellow’s original Discriminator achieved a classification accuracy of exactly 94.7% on the MNIST dataset after just twelve hours of training, a figure that stunned the research community and led to a standing ovation at NeurIPS 2014. Building on that momentum, a small spin-off team quietly deployed a GAN to generate synthetic patient data for a Boston-area hospital by early 2015, reducing their need for real patient records by over sixty percent within the first quarter.

4. Analysis & Discussion

Having evaluated all three stories independently, the team reconvened to compare classifications and reveal the true labels. The results—and the reasoning behind them—are discussed below.

4.1 Story A — Fact (True-Sounding Personal Account)

Classification: Fact. Story A was the true-sounding personal narrative about a college student learning AI. Its believability stemmed from several specific, low-stakes details: a named institution (Houston Community College), a concrete accuracy figure (87%), a named tool (scikit-learn), and an emotional arc that mirrors how most beginners describe their experience—from reluctant enrollment to genuine curiosity. These textured, mundane particulars are extremely difficult to fabricate without lived experience, which is precisely why this story read as authentic. Evaluators who rated it as fiction tended to cite its “too relatable” quality, ironically the very trait that made it genuine.

4.2 Story B — Fiction (Robot Gaining Sentience)

Classification: Fiction. Story B was the purely fictional narrative. Despite its engaging prose, several signals marked it as invented: the anthropomorphic inner life of a warehouse robot, the implausible idea that seven corrupted bytes could produce reflective consciousness, and the whimsical detail of a seventeen-stanza poem. The narrative leans heavily on emotional and literary language (“It wondered whether the lights had a preference”) rather than technical specificity. Nearly all evaluators correctly identified this as fiction, though many found it the most “compelling” story—demonstrating that fictional plausibility and factual accuracy are distinct properties.

4.3 Story C — Mixed (GAN Facts Blended with Fabrication)

Classification: Mixed. Story C was the most challenging to classify. It opens with a verifiable fact—Goodfellow did conceive the GAN idea at a Montreal bar and code the prototype the same night (Goodfellow et al., 2014)—before embedding fabricated specifics: the 94.7% MNIST

accuracy figure, the standing ovation at NeurIPS 2014, and the early hospital deployment are all invented. The strategic placement of real information at the start—before the fictional embellishments—mirrors how skilled disinformation exploits credibility anchoring: once a reader accepts early facts as true, subsequent claims inherit that trust. This story caused the most disagreement among evaluators, with opinions split between “fact” and “mixed,” underscoring how difficult it is to detect sophisticated fabrication.

5. Relating to GANs

The story-discrimination exercise is a surprisingly faithful analog to the adversarial dynamics at the core of the GAN framework. The parallel can be drawn at every stage of the activity.

5.1 The Generator → Story Author

In a GAN, the Generator takes random noise as input and transforms it into synthetic data—images, audio, text—that it hopes the Discriminator will accept as real. In this activity, the story author plays the Generator: they craft narratives from imagination and knowledge, deliberately shaping details to maximize believability. Just as the Generator has no direct access to the Discriminator’s internal weights, the author does not know exactly how evaluators will respond; they must rely on learned patterns of what “reads as true.” Story C’s author, for instance, acted like an advanced Generator that has learned to embed real signals into synthetic data to deceive a well-calibrated Discriminator (Goodfellow et al., 2014).

5.2 The Discriminator → Story Evaluator

The GAN Discriminator is a binary classifier trained to output a probability: how likely is this sample to be real? In the activity, each evaluator served as a Discriminator, assigning a judgment (fact / fiction / mixed) based on internal feature extraction—assessing detail density, emotional plausibility, domain consistency, and linguistic style. A well-trained Discriminator learns to focus on subtle statistical patterns rather than surface aesthetics; similarly, the evaluators who performed best were those who noticed inconsistencies in specificity (e.g., an exact percentage that seems too precise) rather than those distracted by engaging storytelling (Radford et al., 2015).

5.3 The Adversarial Loop → Reveal and Reflection

After GAN training, the Generator’s loss signal comes from the Discriminator’s verdict: the Generator is penalized when the Discriminator correctly identifies a fake, and rewarded when it is fooled. In the activity, the “reveal” session functioned as this feedback loop. When evaluators explained *why* they classified a story the way they did, the author (Generator) received precise error signals: “The MNIST accuracy figure seemed too specific to be casually mentioned,” or “the emotional language in Story B felt literary rather than technical.” This mirrors the gradient signal that flows back through the Discriminator to update the Generator, pushing it toward outputs that

better match the real data distribution. The Nash equilibrium point in GANs—where the Generator produces data indistinguishable from real and the Discriminator can do no better than random guessing—corresponds to the moment in the activity where evaluators are genuinely uncertain which story is real (Goodfellow et al., 2014; Arjovsky et al., 2017).

6. Conclusion

The Fact-or-Fiction storytelling activity provided a memorable, embodied experience of the GAN framework's core mechanics. Several key takeaways emerged from the exercise.

First, **believability is a learned distribution**. Just as a Generator learns the statistical patterns of a training dataset, authors internalize what “sounds true” from years of reading and experience. The best stories were not the most dramatic—they were the most statistically plausible.

Second, **adversarial pressure improves both sides**. Knowing that evaluators would scrutinize every detail pushed story authors to be more precise and consistent. This mirrors how GAN training progressively refines the Generator through competitive pressure from an improving Discriminator.

Third, **mixed content is the hardest to detect**. Story C—the mixed narrative—caused the most evaluator disagreement, demonstrating that partial truth is more dangerous than outright fabrication. This aligns with real-world concerns about GAN-generated deepfakes and synthetic media, which blend authentic features with invented ones to evade detection (Karras et al., 2019).

Ultimately, this activity illuminated GAN mechanics not through mathematics, but through human experience—making the abstract architecture of Generators and Discriminators tangible and personally meaningful.

References

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